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Department of Mathematics
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Math 101 Major Exam I Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Which of the following statements is always true.

(a) If f is continuous on (a,b), $f(a) \neq f(b)$ and k is any number between $f(a)$ and $f(b)$, then there exists a number $c \in (a,b)$ such that $f(c) = k$

(b) $\lim_{x\to c} f(x) = L$ means that for any given positive number q we can find a positive number p such that $|f(x) - L| < q$ whenever $0 < |x - c| < p$

(c) If f and g are functions such that g is continuous at c, then $f(g(x))$ is continuous at c

(d) If f and g are functions such that $\lim_{x\to c} g(x) = L$, then $\lim_{x\to c} f(g(x)) = f(L)$

(e) If f is a function defined on $[a,b]$, $f(a) \neq f(b)$ and k is any number between $f(a)$ and $f(b)$, then there exists a number $c \in (a,b)$ such that $f(c) = k$

2. The slope of the tangent line to the graph of $f(x) = \sin^2 x$ at $x = \dfrac{\pi}{4}$ is equal to

(a) $\lim_{x\to\frac{\pi}{4}} \frac{\sin^2 x - \frac{1}{2}}{x - \frac{\pi}{4}}$

(b) $\lim_{x\to\frac{\pi}{4}} \frac{\sin^2 x - \frac{\sqrt{2}}{2}}{x - \frac{\pi}{4}}$

(c) $\lim_{x\to\frac{\pi}{4}} \frac{\sin^2 x - \frac{\sqrt{2}}{2}}{x^2 - \frac{\pi}{4}}$

(d) $\lim_{x\to\frac{\pi}{4}} \frac{\sin^2 x - \frac{1}{2}}{x^2 - \frac{\pi^2}{16}}$

(e) $\lim_{x\to\frac{\pi}{4}} \frac{\sin x - \frac{\sqrt{2}}{2}}{x - \frac{\pi}{4}}$

3. If $x^3 - x + 3 \leq 7x + \dfrac{f(x)}{x} \leq 3x^2 + x - 3$, then $\lim_{x\to 3} f(x) =$

(a) 18

(b) 6

(c) 27

(d) 24

(e) 21

4. The number of discontinuities of the function $f(x) = x\lfloor x \rfloor$ on the interval $\left(-\frac{5}{2}, \frac{5}{2}\right)$ is

(a) 0

(b) 6

(c) 2

(d) 3

(e) 4

5. If $\lim_{x\to 0} \frac{\sqrt{2}-\sqrt{ax^2+b}}{x^2} = \frac{1}{\sqrt{2}}$, then $a + b =$

(a) 4

(b) 2

(c) -2

(d) 6

(e) 0

6. $\lim_{x\to 0} \frac{\sin\frac{\pi}{3}\sin 2x+\frac{1}{2}\cos x-\frac{1}{2}}{x} =$

(a) 1

(b) $\sqrt{3}$

(c) $\sqrt{3}-1$

(d) $-\sqrt{3}$

(e) -1

7. If a ball is thrown into the air with a velocity of 4 m/s, its height (in meters) t seconds later is given by $y = 4t - 4.9t^2$. The average velocity for the time period from $t = 1$ to $t = 3$ is

(a) -13.7

(b) 13.7

(c) 18.6

(d) -15.6

(e) 17.7

8. The value of c for which the function

$$f(x) = \begin{cases} c^2 + cx^2, & x < 2 \\ 6c, & x = 2 \\ cx + x^3, & x > 2 \end{cases}$$

has a removable discontinuity is

(a) -2

(b) -4

(c) 0

(d) 4

(e) 2

9. The value of k such that the line $y = 4x - 3$ is tangent to the graph of $f(x) = kx^2$ is

(a) $\frac{1}{3}$

(b) $\frac{3}{4}$

(c) $\frac{4}{3}$

(d) 2

(e) $\frac{1}{2}$

10. For the function $f(x) = \frac{1}{1-x}$, the largest value of δ such that $|f(x) + 1| < 0.1$ whenever $0 < |x - 2| < \delta$ is equal to

(a) $\frac{1}{10}$

(b) $\frac{1}{12}$

(c) $\frac{1}{11}$

(d) $\frac{1}{9}$

(e) $\frac{1}{8}$

11. $\lim_{x\to 1} \frac{x^3-1}{x^2-2x+1} =$

(a) $-\infty$

(b) DNE

(c) 0

(d) 1

(e) ∞

12. The number of points on the graph of $f(x) = x^3 + 4$ having a tangent line parallel to the line $3x - y + 2 = 0$ is

(a) 2

(b) 4

(c) 3

(d) 0

(e) 1

13. If $f(x) = -xe^x$, then $f^{(51)}(1) =$

(a) $-51e$

(b) $-51e^2$

(c) $-50e$

(d) $-52e^2$

(e) $-52e$

14. The function $f(x) = \dfrac{\sin(x^2-1)}{x(x+1)}$

(a) has one vertical asymptote

(b) has $\lim_{x\to -1} f(x) = \infty$

(c) has $\lim_{x\to(-1)^-} f(x) \neq \lim_{x\to(-1)^+} f(x)$

(d) is continuous on $\mathbb{R}$

(e) has two vertical asymptotes

15. $\lim_{x\to 0} \frac{1+x}{1+e^{1/x}} =$

(a) 0

(b) ∞

(c) DNE

(d) 2

(e) 1

16. Let $f(x) = \begin{cases} a + \tan \frac{\pi x}{4}, & |x| < 1 \\ bx, & |x| \geq 1 \end{cases}$. If f is continuous on $\mathbb{R}$, then $a^2 + b^2 =$

(a) 8

(b) 5

(c) 1

(d) 4

(e) 10

17. If $f(x) = \dfrac{1+\cos x}{1-\cos x}$, then $f'\left(\dfrac{\pi}{2}\right) =$

(a) $\sqrt{3}$

(b) 2

(c) $-\sqrt{3}$

(d) 0

(e) -2

18. A horizontal tangent line to the graph C of the function $f(x) = \dfrac{x^2}{x-1}$ at a point on C is

(a) $y = -2$

(b) $y = 2$

(c) $y = 4$

(d) $y = -1$

(e) $y = -4$

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	Only the epsilon-delta definition of a limit is always true; the others omit hypotheses.
2	**(a)**	$f'(c) = \lim \frac{f(x)-f(c)}{x-c}$ with $f(\pi/4) = \frac{1}{2}$.
3	**(a)**	Both bounds $\to 27$, so $21 + \frac{L}{3} = 27$ and $L = 18$.
4	**(e)**	Right limit n^2 vs left limit $n(n-1)$ differ unless $n = 0$: discontinuous at $\pm 1, \pm 2$, so 4.
5	**(e)**	$b = 2$ (numerator $\to 0$), conjugate gives $\frac{-a}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$, $a = -2$; $a + b = 0$.
6	**(b)**	$\sqrt{3} \cdot \frac{\sin 2x}{2x} \to \sqrt{3}$ and $\frac{1-\cos x}{x} \to 0$, so the limit is $\sqrt{3}$.
7	**(d)**	$\frac{y(3)-y(1)}{2} = \frac{-32.1+0.9}{2} = -15.6$.
8	**(b)**	Limits agree for $c = -4, 2$; $c = 2$ gives continuity, $c = -4$ gives limit $0 \neq f(2) = -24$.
9	**(c)**	$2ka = 4$ and $ka^2 = 4a - 3$ give $a = \frac{3}{2}$, so $k = \frac{4}{3}$.
10	**(c)**	Interval $\frac{21}{11} < x < \frac{19}{9}$; $\delta = \min\{\frac{1}{11}, \frac{1}{9}\} = \frac{1}{11}$.
11	**(b)**	$\frac{x^2+x+1}{x-1} \to +\infty$ from the right and $-\infty$ from the left: DNE.
12	**(a)**	$f'(x) = 3x^2 = 3 \Rightarrow x = \pm 1$: two points.
13	**(e)**	$f^{(n)}(x) = -(n + x)e^x$, so $f^{(51)}(1) = -52e$.
14	**(a)**	At $x = -1$ the limit is 2 (removable); at $x = 0$ $f \to \pm\infty$: one vertical asymptote.
15	**(c)**	Left limit = 1, right limit = 0, so the limit does not exist.
16	**(c)**	$a + 1 = b$ and $a - 1 = -b$ give $a = 0$, $b = 1$, so $a^2 + b^2 = 1$.
17	**(e)**	$f'(x) = \frac{-2\sin x}{(1-\cos x)^2}$, so $f'(\pi/2) = -2$.
18	**(c)**	$f' = \frac{x^2-2x}{(x-1)^2} = 0$ at $x = 0, 2$; tangent lines $y = 0$ and $y = 4$.